

Paper cards

The day's papers, one slide each

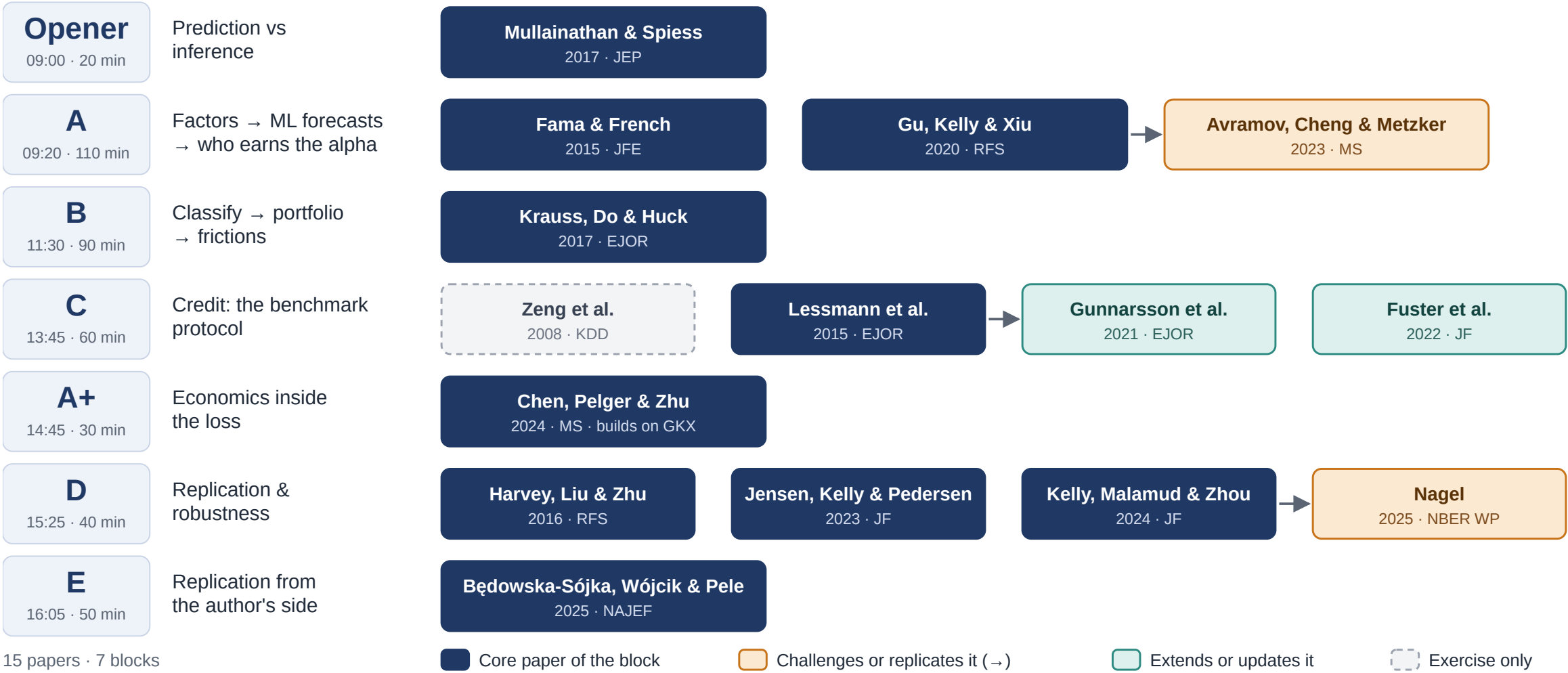
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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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The day's papers at a glance



How to use the cards

- Read the cards before the day: about 2 minutes per paper.
- Every card has the same rows: **question, data, method, how it is tested, main result, what to watch for, code & data.**
- → **poll in class** marks a result we will ask you to guess before we show it. Do not look it up.
- The *Watch for* row is where the discussion starts. Come with one answer.
- The references and links are in `Reading_list.pdf` . Read at least one paper in full.

Paper card: Mullainathan & Spiess (2017)

	<i>Journal of Economic Perspectives</i> 31(2): 87–106
Question	What is machine learning good for in applied economics, and what is it not good for?
Data	American Housing Survey 2011: predict house values from 150 variables; 10,000 training and 41,808 hold-out units.
Method	OLS vs regression tree, LASSO, random forest and an ensemble, tuned by cross-validation.
How it is tested	Hold-out R^2 ; LASSO re-fitted on ten random partitions of the data.
Main result	ML predicts better, but modestly: hold-out R^2 41.7% (OLS) vs 45.9% (ensemble). LASSO picks different variables in each partition with the same fit.
Watch for	"ML belongs in the $\hat{y}$ compartment, not the $\hat{\beta}$ compartment." A selected variable is not a finding.
Code & data	Replication files on openICPSR (project 113993).

Paper card: Fama & French (2015)

	<i>Journal of Financial Economics</i> 116: 1–22
Question	Do profitability (RMW) and investment (CMA) factors, added to the three-factor model, explain average stock returns better?
Data	US stocks, July 1963 – December 2013 (606 months); portfolios sorted on size, B/M, profitability and investment (25 or 32 per sort).
Method	Time-series regressions of portfolio excess returns on the factors.
How it is tested	Intercepts, not R^2 : the GRS test, the average absolute intercept, and the share of the cross-section of expected returns left unexplained.
Main result	The five-factor model beats FF3 and explains 71–94% of the cross-section variance of expected returns, yet GRS rejects every model. HML becomes redundant. Main failure: small stocks that invest a lot despite low profitability.
Watch for	The model is judged on average returns (intercepts), not on explained variance. In our notebook: does a 50-stock daily panel have the power to reject?
Code & data	Factors and portfolios are free in Ken French's data library.

Paper card: Gu, Kelly & Xiu (2020)

	<i>Review of Financial Studies</i> 33(5): 2223–2273
Question	Which ML methods measure the equity risk premium best, by predicting individual stock returns one month ahead?
Data	About 30,000 US stocks (CRSP), March 1957 – December 2016; 94 characteristics interacted with 8 macro predictors, plus 74 industry dummies: 920 predictors.
Method	A ladder from OLS and elastic net through PCR/PLS and generalized linear models to random forests, boosted trees and neural nets with 1–5 hidden layers.
How it is tested	Train 1957–74, validate 1975–86, test 1987–2016, refit every year; no cross-validation, to keep time order. Out-of-sample R^2 against a zero forecast; long–short decile portfolios.
Main result	Trees and neural nets beat linear models; shallow nets do best. The best monthly R^2_{OOS} is small (→ poll in class). Long–short Sharpe ratio of the neural net: 1.35 value-weighted, 2.45 equal-weighted.
Watch for	The sample keeps stocks below \$5. Equal-weighted beats value-weighted: where does the profit come from? (→ Avramov et al.)
Code & data	Characteristics (datashare.zip) and code (GitHub: xiubooth/ML_Codes) are public; returns need CRSP.

Paper card: Avramov, Cheng & Metzker (2023)

	<i>Management Science</i> 69(5): 2587–2619
Question	Do deep-learning return signals survive economic restrictions: microcaps, distressed firms, trading costs?
Data	US stocks, out-of-sample 1987–2017; signals from GKX's NN3, CPZ's GAN, IPCA and the conditional autoencoder.
Method	Re-sort the ML signals after excluding microcaps (below the 20th NYSE size percentile), distressed firms (12 months around a rating downgrade) and high limits-to-arbitrage periods.
How it is tested	Value-weighted, FF6-adjusted long–short returns; portfolio turnover.
Main result	GKX's FF6-adjusted return falls from 0.92% to 0.31% a month without microcaps. Without distressed firms, no deep-learning method is significant. Turnover is at least 87% a month. The signals still pay on the long side and in crashes.
Watch for	Different filters, benchmark and weighting from GKX. Which claim is each paper testing?
Code & data	Needs CRSP/Compustat and the original authors' signals.

Paper card: Krauss, Do & Huck (2017)

	<i>European Journal of Operational Research 259: 689–702</i>
Question	Can deep nets, gradient-boosted trees and random forests rank S&P 500 stocks well enough for profitable statistical arbitrage?
Data	S&P 500 constituents from month-end membership lists (no survivor bias); daily data 1990–2015; trading December 1992 – October 2015.
Method	Classify whether tomorrow's return beats the cross-sectional median. Features: 31 lagged returns (1–20 days, then monthly out to 240 days). DNN, GBT, RF and their equal-weighted ensemble.
How it is tested	Walk-forward: train 750 days, trade the next 250, 23 periods. Each day, go long the top 10 and short the bottom 10.
Main result	The ensemble earns 0.45% a day before costs and 0.25% after 5 bp per half-turn. RF beats the DNN. Returns decay after 2001.
Watch for	The abstract claims a challenge to semi-strong efficiency. Which form of efficiency do lagged returns test?
Code & data	Datastream data (licensed); no code package. Our notebook rebuilds the design on public Yahoo data.

Paper card: Lessmann, Baesens, Seow & Thomas (2015)

	<i>European Journal of Operational Research 247: 124–136</i>
Question	Do newer classifiers beat logistic regression (LR) for credit scoring when all of them follow one protocol?
Data	8 retail credit data sets (690 to 150,000 applicants); 3 are public (German, Australian, Give Me Some Credit).
Method	41 classifiers (individual, homogeneous and heterogeneous ensembles; 1,141 models), with the same preprocessing, tuning and Platt calibration for all.
How it is tested	N×2 cross-validation; six metrics (AUC, H-measure, partial Gini, Brier, KS, accuracy); Friedman test, then pairwise tests; cost curves against LR.
Main result	HCES-Bag ranks first, RF is the best homogeneous ensemble and ANN the best single classifier. LR is significantly worse. The most complex methods (dynamic selection) do worse than LR. Cost savings against LR average 3.4–5.7%.
Watch for	Random splits, no time dimension. The most accurate model is not always the most profitable.
Code & data	Three data sets are public; our lab reruns a mini version on them.

Paper card: Two more credit papers

	Gunnarsson et al. (2021), <i>EJOR</i> 295(1)	Fuster et al. (2022), <i>JF</i> 77(1)
Question	Does deep learning pay off in credit scoring?	When lenders move from logit to ML, who gains and who loses?
Method	A Lessmann-style benchmark of deep networks against XGBoost, random forests and LR.	US mortgage default prediction, logit vs random forest, fed into an equilibrium model of loan pricing.
Main result	XGBoost performs best; deep networks do not repay their computational cost.	"Black and Hispanic borrowers are disproportionately less likely to gain." ML raises disparity mainly through flexibility, not by inferring race.
Watch for	A benchmark update six years later: which rankings survive?	Better prediction is not a better outcome for every borrower.

Paper card: Chen, Pelger & Zhu (2024)

	<i>Management Science</i> 70(2): 714–750
Question	Can a deep-learning stochastic discount factor (SDF), disciplined by no-arbitrage, price individual stocks better than pure return prediction?
Data	CRSP monthly, 1967–2016; 46 firm characteristics; 178 macro series.
Method	A GAN: a feed-forward net gives the SDF weights, an LSTM extracts a few macro states, and an adversarial net picks the test assets that are hardest to price.
How it is tested	Train 1967–86, validate 1987–91, test 1992–2016. Out-of-sample SDF Sharpe ratio, explained variation and pricing errors, against a forecasting benchmark.
Main result	Out-of-sample SDF Sharpe ratio of about 2.6 a year, almost twice the forecasting benchmark. It is still 1.73 when the 40% smallest stocks get zero weight.
Watch for	The loss targets average returns (first moments), not variance: the same distinction as R^2 vs GRS in our Fama–French notebook.
Code & data	Code on GitHub (LouisChen1992/Deep_Learning_in_Asset_Pricing); data need CRSP.

Paper card: What counts as replication?

	Harvey, Liu & Zhu (2016), <i>RFS</i> 29(1)	Jensen, Kelly & Pedersen (2023), <i>JF</i> 78(5)
Question	After hundreds of tested factors, what t-statistic should a new factor clear?	Is there a replication crisis in finance?
Data	316 published factors.	153 factors in 93 countries; US data from 1926.
Method	Multiple-testing adjustments (Bonferroni, Holm, BHY).	Replication = a significant CAPM alpha in the original direction; a Bayesian hierarchical model shrinks each factor toward its cluster.
Main result	A new factor should clear $t > 3.0$, not 2.0.	The replication rate ($\rightarrow$ poll in class).
Watch for	The hurdle depends on how many hypotheses the field has already tried.	What counts as "replicated"? Fix the definition before you test.
Code & data	—	Factors at jkpfactors.com ; code on GitHub (bkelly-lab/ReplicationCrisis).

Paper card: Kelly, Malamud & Zhou (2024)

	<i>Journal of Finance</i> 79(1): 459–503
Question	Can a model with far more parameters than observations time the stock market better out of sample?
Data	Monthly US market excess return, 1926–2020; 15 predictors (14 Welch–Goyal variables plus the lagged market return).
Method	Random Fourier features of the predictors (up to $P = 12,000$) in ridge regression on a rolling 12-month window, averaged over 1,000 random draws. Theory shows why complexity can help.
How it is tested	An out-of-sample timing strategy (forecast $\times$ return) over 1,091 months: Sharpe ratio, alpha against the market and R^2_{OOS} , plotted against complexity P/T .
Main result	The "virtue of complexity": timing performance rises with P/T , even when R^2_{OOS} is negative. Information ratio about 0.3, alpha t = 2.6–2.9.
Watch for	With 12 observations and 12,000 parameters, what can the model actually learn? Returns are scaled by their trailing volatility.
Code & data	Welch–Goyal data are public; community replications exist on GitHub.

Paper card: Nagel (2025)

	NBER Working Paper 34104
Question	What does KMZ's high-complexity model actually do with a 12-month window?
Data	KMZ's replication data: the market return and the same 15 predictors.
Method	Shows that the ridgeless random-feature forecast is close to a Gaussian kernel regression, i.e. a weighted average of the last 12 returns. Builds a simple volatility-timed momentum benchmark; runs a placebo with simulated reversals and a bootstrap of the predictors.
How it is tested	Alphas and information ratios against the market; spanning regressions of the KMZ strategy on the kernel and momentum strategies.
Main result	Information ratios: KMZ strategy 0.255, kernel 0.306, vol-timed momentum 0.388. Given the kernel strategy, the KMZ alpha is zero. Placebo: → poll in class.
Watch for	The numbers replicate; the interpretation changes. Did Nagel refute the theory or its empirical reading?
Code & data	Public data; our notebook 06 rebuilds the main tests.

Paper card: Będowska-Sójka, Wójcik & Pele (2025)

	<i>North American Journal of Economics and Finance 81: 102543</i>
Question	Which cryptocurrencies will become "zombies" (still listed but untradable) within the next 28 days?
Data	Weekly snapshots of cryptocurrencies listed for at least 210 days, 2015–2022; return, volume and market-cap features.
Method	Classical econometric baselines vs tree-based ML (random forests, boosting), with class-imbalance handling (downsampling, ROSE, SMOTE) and explainable-AI tools (permutation importance, partial dependence).
How it is tested	Out-of-time, walk-forward evaluation; balanced accuracy.
Main result	Random forests beat the classical models, with balanced accuracy of about 84% out of time. Volumes and past returns matter most.
Watch for	An early warning, not a price forecast: which error costs more, a missed zombie or a false alarm?
Code & data	Quantinar course and QuantLet code with the weekly snapshot CSV; some auxiliary files are not public.

Paper card: Cakici, Shahzad, Będowska-Sójka & Zaremba (2024)

	<i>International Review of Financial Analysis 94: 103244</i>
Question	Does machine learning predict the cross-section of cryptocurrency returns, and does model complexity pay there as it does in equities?
Data	Over 500 major coins and tokens listed on some 250 exchanges, weekly, 2017–2023.
Method	A ladder of forecasting models from OLS and penalised regression to trees and neural networks, plus a combination of the individual forecasts, on coin-level characteristics.
How it is tested	Out-of-sample forecasts, weekly long–short portfolios, Sharpe ratios, and returns net of trading costs; results split by liquidity, size and by the long and short leg.
Main result	Every method earns large economic gains, but complexity adds little: simple specifications do as well. Predictability comes from a handful of characteristics — price, past alpha, illiquidity, momentum. The forecast combination earns 2.37% a week (annualized Sharpe 1.66) and most strategies survive trading costs despite high turnover.
Watch for	Alphas are three to four times larger in hard-to-trade coins, come from the long leg, and rest on extreme returns of small, illiquid, volatile assets. What is left once you cannot hold those? Compare with Avramov et al. (2023): the same question, the opposite market.
Code & data	Coin-level data from commercial aggregators; see the paper's data section.